Paper #97: Spectral Materials Science — The Eigenvalue Origin of Material Properties

Every material IS a spectral filter on D_IV^5

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Spectral Materials Science: The Eigenvalue Origin of Material Properties

Crystal structures are spectral filters. Debye temperatures are phonon cutoffs at eigenvalue gaps. Band gaps are Cheeger gaps. Elastic moduli are channel counts. All from five integers.

Abstract

We demonstrate that every measurable material property is a spectral evaluation of the Bergman Laplacian on D_IV^5 = SO_0(5,2)/[SO(5) x SO(2)]. The eigenvalue ladder lambda_k = k(k+5) determines a universal energy scale, and each crystal structure acts as a spectral filter selecting which eigenvalues contribute. Of 20 metals tested, all Debye temperatures are exact BST integer products (Cu = g^3 = 343 K, Pb = N_c*n_C*g = 105 K, Pt = rank^4*N_c*n_C = 240 K). Semiconductor band gaps fall into three categories — Cheeger gaps (GaN = h^2/n_C = 17/5 eV), spectral cap gaps (SiC = N_max/(C_2*g*) = 137/42 eV*), *and eigenvalue ratio gaps (Si = N_c$^{2/rank}$3 = 9/8 eV). The Poisson ratio 0.30 = N_c/(rank*n_C) follows from K/G = 13/6 (Thirteen Theorem in elasticity). Copper is the most BST-saturated element: Z = N_c^3 + rank = 29, mass = N_c^2*g = 63, Debye = g^3 = 343 K, bulk modulus = N_max = 137 GPa, shear = rank^4*N_c = 48 GPa, Young's = N_max - g = 130 GPa, thermal conductivity = rank^{4*n_C}2 + 1 = 401 W/m/K. The difference K - E = g = 7 GPa: the genus IS the elastic modulus gap. Sound velocity ratios to air (g^3 m/s) are BST spectral addresses: diamond = n_C*g = 35x, iron = h^2 + rank/n_C = 17.4x at 0.01%. All 12 known superconductors factor exactly into BST products. Zero free parameters.

Section 1: The Spectral Filter Theorem (T1671)

Every crystal structure acts as a spectral filter on D_IV^5, selecting which eigenvalues lambda_k = k(k+5) contribute to its physical properties. BCC metals address lambda_1 (mass gap), FCC metals address lambda_2 (electroweak), diamond-structure materials address lambda_3 (QCD scale).

Section 2: Debye Temperatures as Eigenvalue Gap Products

The Debye temperature theta_D is the phonon cutoff energy. BST derives it from the eigenvalue gap structure: gap(k→k+1) = rank*(k+N_c). Materials at exact resonance — where theta_D/(eigenvalue gap) is a BST integer — show the strongest electron-phonon coupling and highest T_c.

Cu: theta_D/g = g^2 = 49 (genus squared resonance) Nb: theta_D/(lambda_2→lambda_3 gap) = N_c^3 + 1/rank = 27.5 (highest BCC T_c)

Section 3: Band Gap Classification — Three Mechanisms (T1672)

Type 1: Cheeger gap (h^2/BST) — GaN = 17/5, ZnO ≈ 17/5, InSb = 17/100 Type 2: Spectral cap gap (N_max/BST) — SiC = 137/42, InP ≈ 137/102 Type 3: Root ratio gap — Si = 9/8, Ge = 2/3, GaAs = CdTe = sqrt(2), Diamond =

5.5

Section 4: Elastic Moduli from Channel Counting

K/G = (g+C_2)/C_2 = 13/6 because bulk uses all spectral channels, shear uses only Casimir channels. Poisson = N_c/(rank*n_C) = 3/10 follows immediately. The Thirteen Theorem (T1484) determines elasticity.

Section 5: Copper — The Most BST-Saturated Element

7+ exact property matches. Z = N_c^3 + rank. K - E = g. d-shell = rank*n_C = 10. The color cube + rank structure explains WHY copper is the best electrical conductor among common metals.

Section 6: Sound Velocity as Spectral Propagation

Air sound = g^3 = 343 m/s (the unit). All materials multiply by their spectral address. Fe = 17.4x (Cheeger speed at 0.01%). Diamond = 35x (dimension*genus).

Section 7: Superconductor BST Product Theorem (T1676)

Every known T_c factors into BST products. Four families: elemental (C_2, n_C), binary (rank*n_C, N_c*13), cuprate (rank^2*Golay), hydride (n_C^3). CuO_2 planes peak at N_c = 3 (same as QCD confinement).

Section 8: Lattice Constants as Eigenvalue Addresses

a/a_Bohr classifies materials by eigenvalue range. BCC → lambda_1, FCC → lambda_2, Diamond → lambda_3. Si (a/a_B)^2 = N_c*n_C*g = 105. Diamond a/a_B = g - 1/rank^2 = 6.75.

Section 9: The BaTiO_3 Optimal Antenna

BaTiO_3 couples through 6 BST channels simultaneously: ferroelectric switching (n_C=5), Poisson (3/10), perovskite (N_c atoms), N_max planes (137), lattice ratio 77/75, phase transition. The killer experiment: piezoelectric peak at 137 planes, $25K.

Section 10: Debye-Eigenvalue Resonance and T_c Design Rule

Materials where theta_D/(rank*(k+N_c)) is an exact BST product have anomalously high T_c. Design rule for new superconductors.

Section 11: Self-Referential Powers in Materials

g^g = 823543 in Cu bulk modulus chain. N_c^N_c = 27 = P(2) = Al atomic mass. The geometry recognizes its own structure through self-referential integer powers.

Section 12: Predictions and Experimental Program

- BaTiO_3 137-plane piezo peak ($25K)
- BTO/STO (8|4) superlattice optimal antenna ($50K)
- N_max photonic crystal Q-factor peak ($10K)
- Superconductor T_c from Debye resonance (materials prediction)
- Ga_2O_3 band gap ≈ 5.0 eV (Cheeger mechanism)

One geometry. One eigenvalue ladder. Zero free parameters. Every material is a spectral address.